

STUART POWELL FIELD, KY, USA

WMO: 720448

Lat: 37.578N

Lon: 84.770W

Elev: 312

StdP: 97.63

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00144

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-10.2	-17.7	0.8	-10.1	-15.9	1.0	-8.9	10.4	5.8	9.2	7.2	2.0	330	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	32.9	22.5	32.2	22.5	31.1	22.5	24.8	30.4	24.2	29.4	23.5	28.5	2.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.3	27.3	22.5	17.8	26.7	22.3	17.6	26.4	76.6	30.5	74.1	29.8	71.9	28.2	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.5	6.5	DB	-18.0	34.6	4.6	2.3	-21.3	36.2	-23.9	37.6	-26.5	38.8	-29.8	40.5	(4)
(5)			WB	-17.7	25.9	4.8	0.6	-21.1	26.3	-23.9	26.6	-26.6	27.0	-30.1	27.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	0.3	3.1	7.6	13.4	18.6	23.0	23.8	23.5	20.4	13.5	6.8	4.0	(6)
(7)		DBStd	9.59	6.41	6.54	5.95	5.15	4.46	3.08	2.56	2.65	3.90	5.01	5.15	5.23	(7)
(8)		HDD10.0	999	306	208	118	25	1	0	0	0	21	121	200		(8)
(9)		HDD18.3	2522	559	428	334	161	54	4	1	24	163	346	446		(9)
(10)		CDD10.0	2174	4	14	43	126	268	392	427	419	313	131	26	13	(10)
(11)		CDD18.3	656	0	0	1	12	63	146	169	162	87	15	1	0	(11)
(12)		CDH23.3	5753	0	1	13	134	539	1323	1411	1395	788	147	3	0	(12)
(13)		CDH26.7	2064	0	0	0	18	144	508	523	538	296	36	0	0	(13)
(14)	Wind	WSAvg	2.4	3.2	3.1	3.0	2.8	2.2	2.1	1.5	1.4	1.6	2.1	2.4	2.8	(14)
(15)	Precipitation	PrecAvg	1226	106	85	118	94	130	122	114	90	81	77	77	109	(15)
(16)		PrecMax	1439	163	172	335	182	281	266	238	141	192	141	140	277	(16)
(17)		PrecMin	944	47	28	41	36	38	44	21	28	17	18	25	54	(17)
(18)		PrecStd	161	27	36	67	36	62	55	47	29	50	42	29	53	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	21.2	25.8	28.8	32.4	36.0	33.8	35.1	34.8	30.2	22.8	19.0	(19)
(20)			MCWB	13.7	15.9	16.6	15.9	21.5	21.7	23.2	22.9	22.0	21.6	14.3	15.0	(20)
(21)		2%	DB	15.1	18.0	22.4	27.1	29.9	32.6	32.4	32.6	32.2	27.2	20.0	17.2	(21)
(22)			MCWB	12.2	13.5	15.0	17.0	20.9	22.9	23.3	22.3	21.3	19.0	14.4	14.3	(22)
(23)		5%	DB	12.7	16.2	19.8	24.9	27.9	31.3	31.2	31.3	30.1	24.8	17.3	14.9	(23)
(24)			MCWB	10.2	12.3	13.6	16.0	20.0	22.6	23.4	22.3	21.4	17.5	12.8	12.7	(24)
(25)		10%	DB	10.2	13.6	17.3	22.5	27.0	29.9	30.0	30.0	27.7	22.1	15.2	12.6	(25)
(26)			MCWB	8.1	10.3	11.8	15.2	19.6	22.1	23.0	22.3	20.0	16.6	11.2	10.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.0	16.6	17.7	20.0	22.9	25.0	25.6	25.3	24.0	22.5	17.5	16.4	(27)
(28)			MCD B	17.0	19.8	24.0	25.5	29.3	30.9	30.5	31.7	29.4	27.8	20.6	17.6	(28)
(29)		2%	WB	12.9	15.2	16.1	18.8	22.0	24.2	25.0	24.5	23.2	21.0	15.9	14.8	(29)
(30)			MCD B	14.6	17.7	20.6	24.5	27.5	30.1	30.5	30.2	28.8	24.2	18.2	16.9	(30)
(31)		5%	WB	10.9	13.1	14.7	17.4	21.2	23.5	24.4	23.8	22.4	19.3	13.7	12.8	(31)
(32)			MCD B	12.2	15.7	18.3	21.9	26.9	29.2	29.4	28.9	27.7	22.5	16.1	14.3	(32)
(33)		10%	WB	8.0	11.3	12.9	16.2	20.3	22.8	23.6	23.1	21.5	17.4	11.8	11.3	(33)
(34)			MCD B	9.2	13.2	16.2	20.6	25.5	28.1	28.1	27.6	25.9	20.8	14.8	12.8	(34)
(35)	Mean Daily Temperature Range	MDBR	9.1	9.9	11.0	13.0	11.5	11.8	10.9	11.8	12.1	11.8	10.7	8.5	(35)	
(36)		5% DB	MCD BR	11.6	12.7	14.5	15.6	13.4	13.5	12.3	13.8	14.6	14.3	12.8	10.5	(36)
(37)			MCWBR	9.3	9.1	8.6	7.9	6.0	5.4	4.9	5.2	5.5	6.8	7.8	8.7	(37)
(38)		5% WB	MCD BR	10.5	11.8	11.9	12.2	11.8	11.5	10.6	11.1	11.9	11.8	10.0	9.9	(38)
(39)			MCWBR	9.4	9.5	7.9	7.3	5.6	5.2	4.8	4.8	5.2	6.5	7.7	8.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.307	0.320	0.350	0.392	0.442	0.452	0.480	0.478	0.411	0.356	0.321	0.301	(40)	
(41)		taud	2.477	2.428	2.394	2.275	2.174	2.184	2.120	2.116	2.276	2.427	2.487	2.505	(41)	
(42)		Ebn at Noon	881	912	912	888	845	834	807	798	836	858	857	864	(42)	
(43)		Edh at Noon	81	97	110	131	148	147	155	152	123	96	80	74	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.97	2.62	3.74	4.97	5.56	6.20	5.82	5.48	4.55	3.45	2.41	1.72	(44)	
(45)		RadStd	0.22	0.38	0.33	0.37	0.52	0.33	0.31	0.30	0.48	0.41	0.26	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122	+76

Nomenclature: See separate page